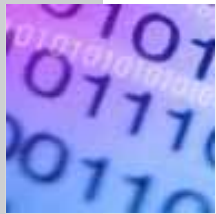


EPCglobal Overview

Nicholas Fergusson / EPCglobal Inc

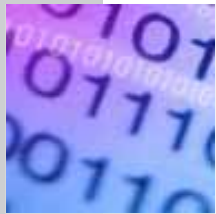
Seoul, South Korea
January 20th, 2006



The Global Footprint of EAN and UCC now GS1



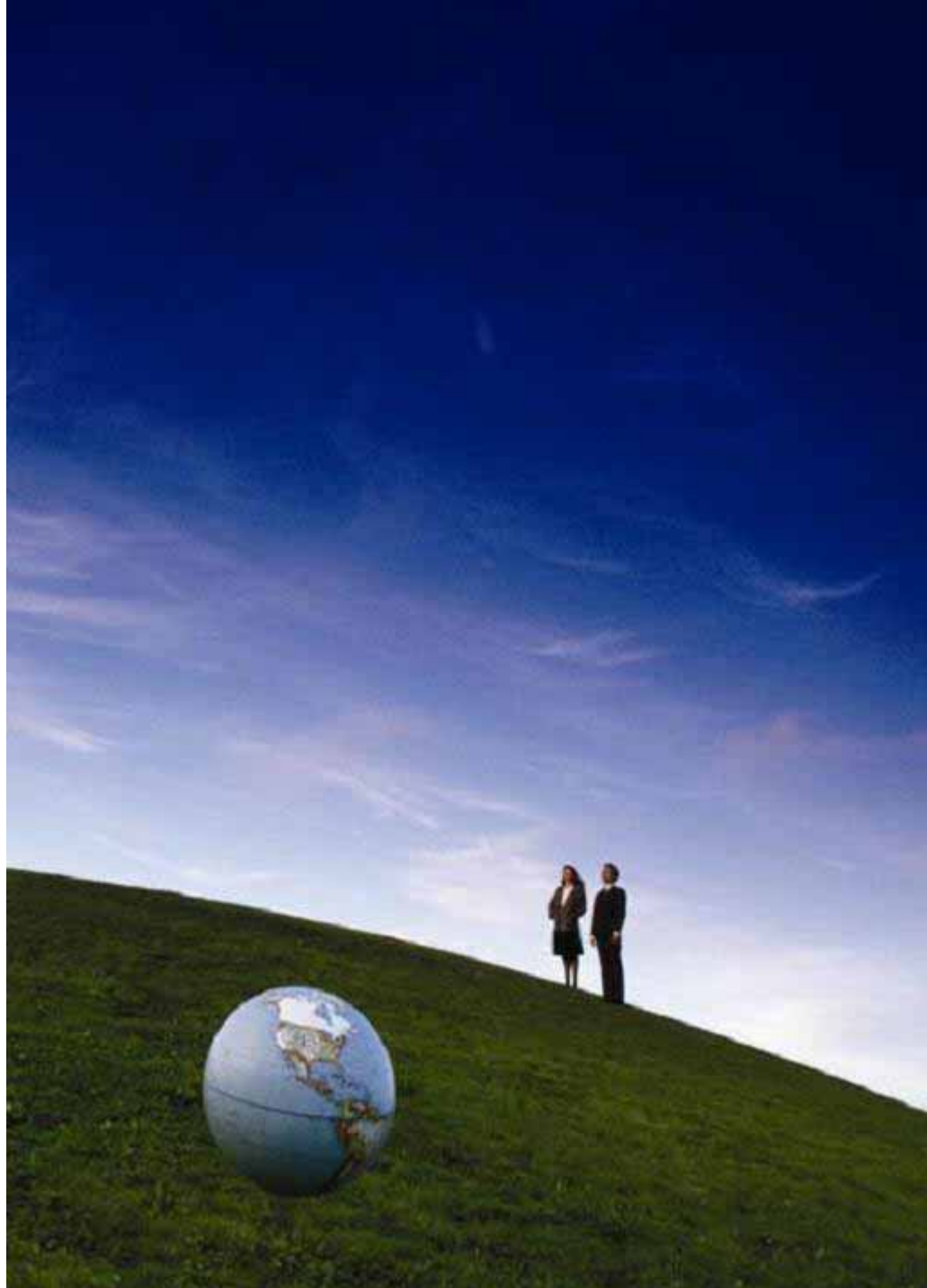
100+ GS1 Member Organizations • One Million Members • Over 100 Countries
Not for Profit Organizations • **Head Office, Brussels, Belgium**

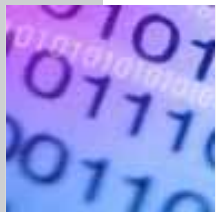


EAN and UCC is now GS1

Single global organization

A product portfolio for
today's business needs and
for the future





GS1: a diversified portfolio

- GS1 has a full portfolio of products and solutions



Global standards for automatic identification
Rapid and accurate item, asset or location identification



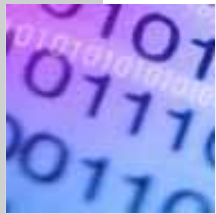
Global standards for electronic business messaging
Rapid, efficient & accurate business data exchange



The environment for global data synchronisation
Standardised, reliable data for effective business transactions

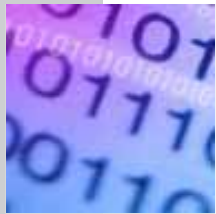


Global standards for RFID-based identification
More accurate, immediate and cost effective visibility of information



EPCglobal Heritage

- Six year research programme supported by over 100 organizations
- End 2003 – responsibility for adoption and standards transferred to EAN / UCC.....now GS1
- EPCglobal created
- Part of the GS1 organization. GS1 covers 1.1million companies, 140 countries. Expert organization – over 30 years experience.

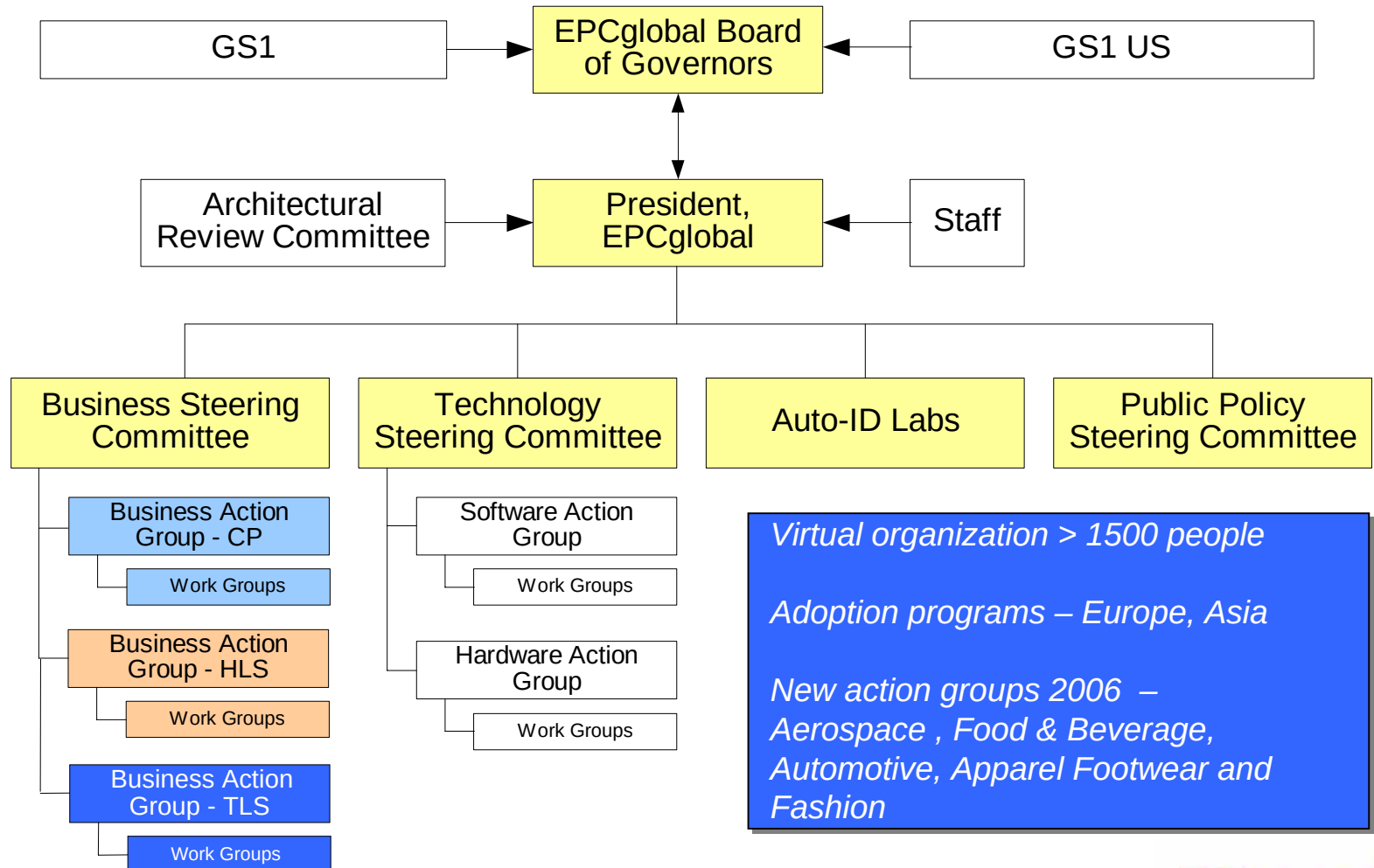


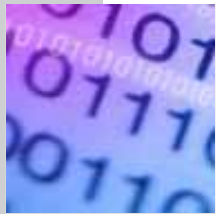
EPCglobal Purpose

1. *Global* leadership in developing and promoting *multi-industry, user driven standards* for *collaborative commerce* utilising the *EPC*
2. Deliver added value to our customers and stakeholders
3. Drive global, multi-industry adoption of EPC via the GS1 Member Organizations
4. Act as trusted authority on technical standards relating to use of EPC
5. Effectively manage public policy issues relevant to the use of the EPC

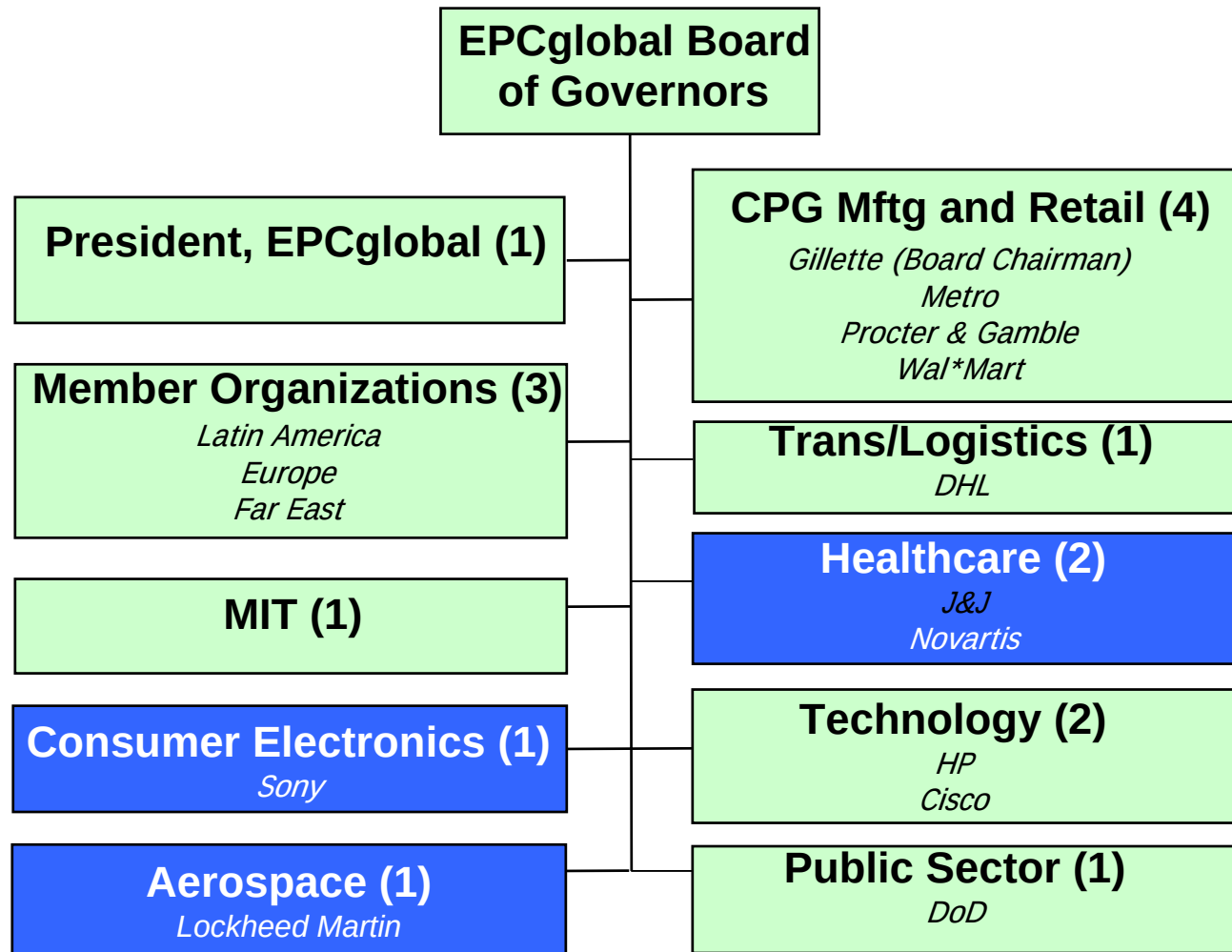


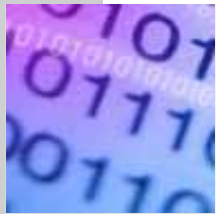
EPCglobal Organization





EPCglobal Board of Governors





Our research partners: Auto ID Labs

Cambridge - Cambridge, UK

MIT – Cambridge, US

Adelaide - Australia

St. Gallen – Switzerland

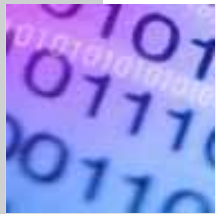
Fudan – China

Keio – Japan

ICU – south Korea



- Low cost, high performance RFID tags & readers enable the EPC vision
- Tags are simple: Most information held in the network
- Serialised items can be individually tracked across traditional trading boundaries in a standardized fashion with unprecedented automation
- The Auto-ID Labs architect the Internet of Things. Their research is global, relevant (partner of EPCglobal), long-term, and cross-discipline

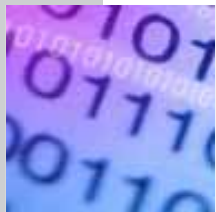


Significant growth in all regions

- EPC global is only two years old!

	Jun 2004	Dec 2005	% Increase
Asia	21	148	705%
North America	132	451	342%
Europe	36	110	306%
ME & Africa	2	8	400%
Latin America	0	16	-
Total	191	733	384%

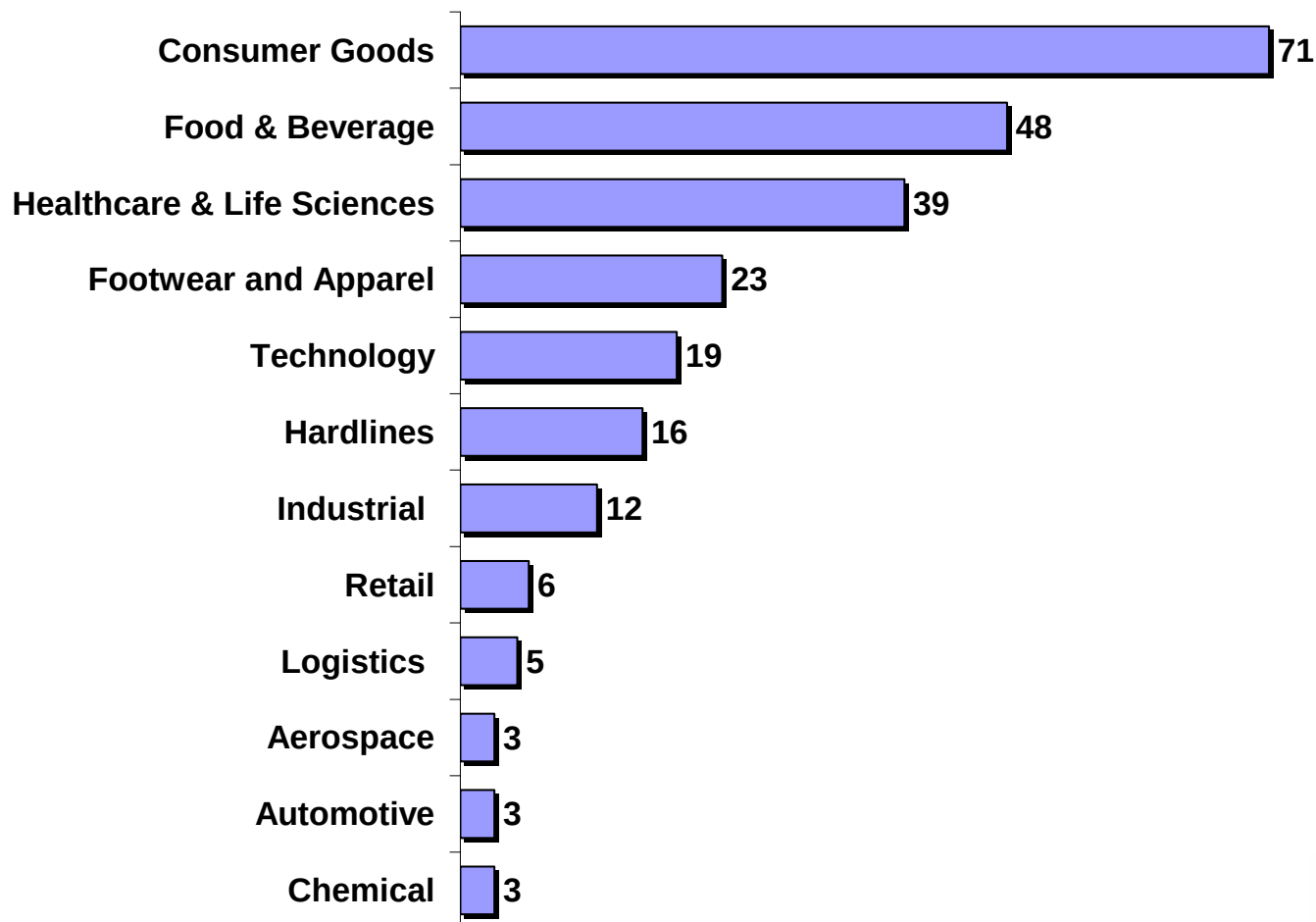
- Note: End 2004 = 391

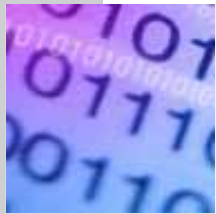


Membership growth in all sectors

US example

- Momentum is developing rapidly in new sectors





Why are global standards important?

- Reduce complexity
 - Within organizations
 - Between organizations
 - For H/W and S/W production and purchase decisions
- Reduce cost
 - Implementation
 - H/W, S/W and Integration
- Facilitate trading partner collaboration
- Allow organizations to focus more on how to use the information than how to get information



Security Specifications

ALE F&C

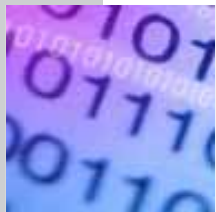
Application Program Interface (API)

Tag Data Translation
Reader Management
Reader Protocols

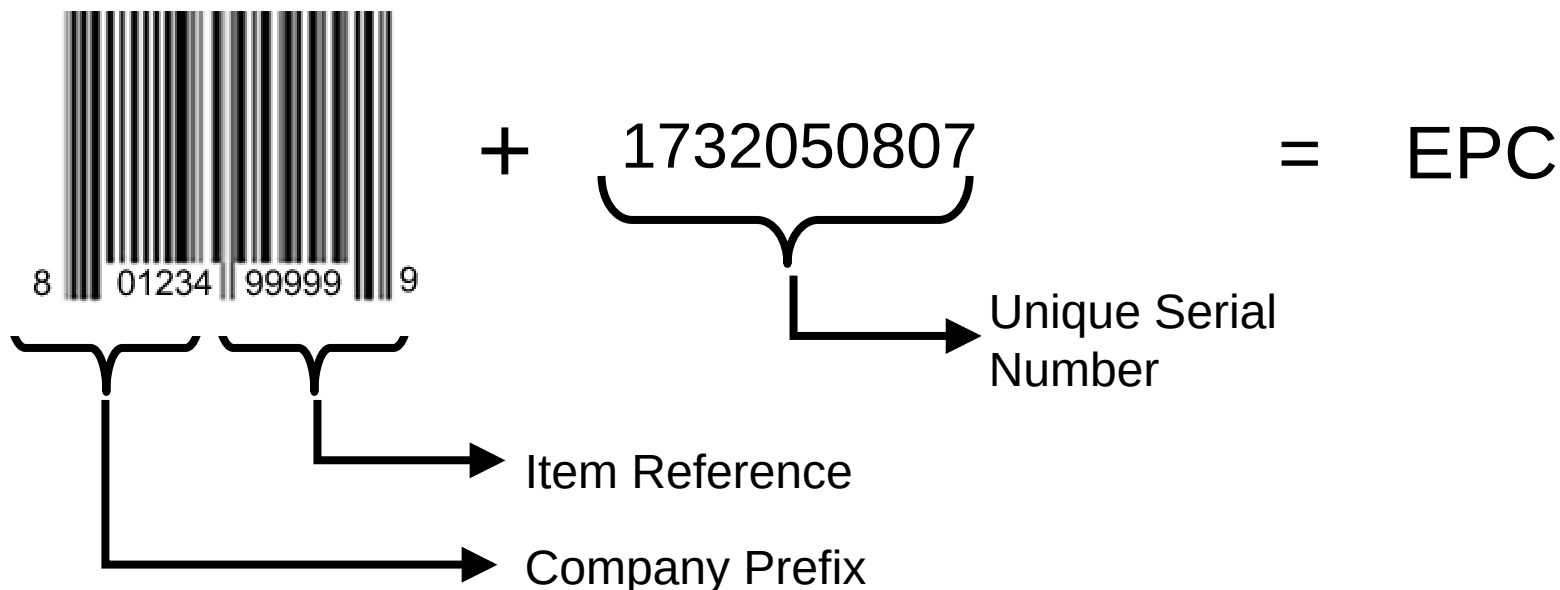
GEN 2 AIP

Tag Data Standards

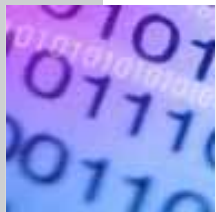
EPCIS protocols



Electronic Product Code

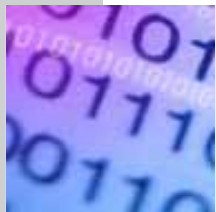


- Every item has distinct serial number
- Capacity for 200 billion serial numbers per item class (on 96-bit tag)
- New business processes based on tracking individual things



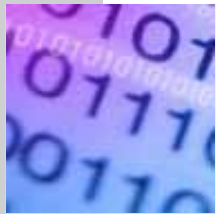
EPCglobal Standards

Standard	Status	Comments
Tag Data Standard	Version 1.27 Ratified	Replaces version 1.24 Under development 1.3
G2 Air Interface	Version 1.09 Ratified	Certification Completed
Reader Protocol	Under Development 1.1	Under Development 1.2
Tag Data Translation	Under Development 1.0	
Reader Management	Under Development 1.0	
Filtering and Collection (ALE)	Version 1.0 Ratified	Under Development 1.1
EPC Information Services	Under Development 1.0	
Object Naming Service	Version 1.0 Ratified	
Security (EPC IS Security Profile)	Under Development 1.0	



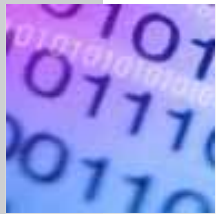
UHF Class 1 Gen 2 Features

Requirement	Gen2 Capability
Global Regulatory Compliance	Europe, North America, Japan, etc.
Operation in Noisy Environments	Multiple Sessions, Dense Reader Modes
Fast Operation	> 1600 tags/sec USA, 600 tags/sec Europe
Privacy Protection	EPC code not broadcasted, 32-Bit Kill Password
Improved Accuracy	Elimination of "Ghost Reads", Adaptive Protocols
Memory <i>Write</i> Capability	> 7 tag/second write rate, Optional User Memory
Group Searches & Filtering	Flexible <i>Select</i> Command
Low Cost	Multi-Vendor Availability
Flexibility	Tolerates Identical EPC numbers & Multiple EPCs
Certified products	Currently Available (starting in Q1 2005)



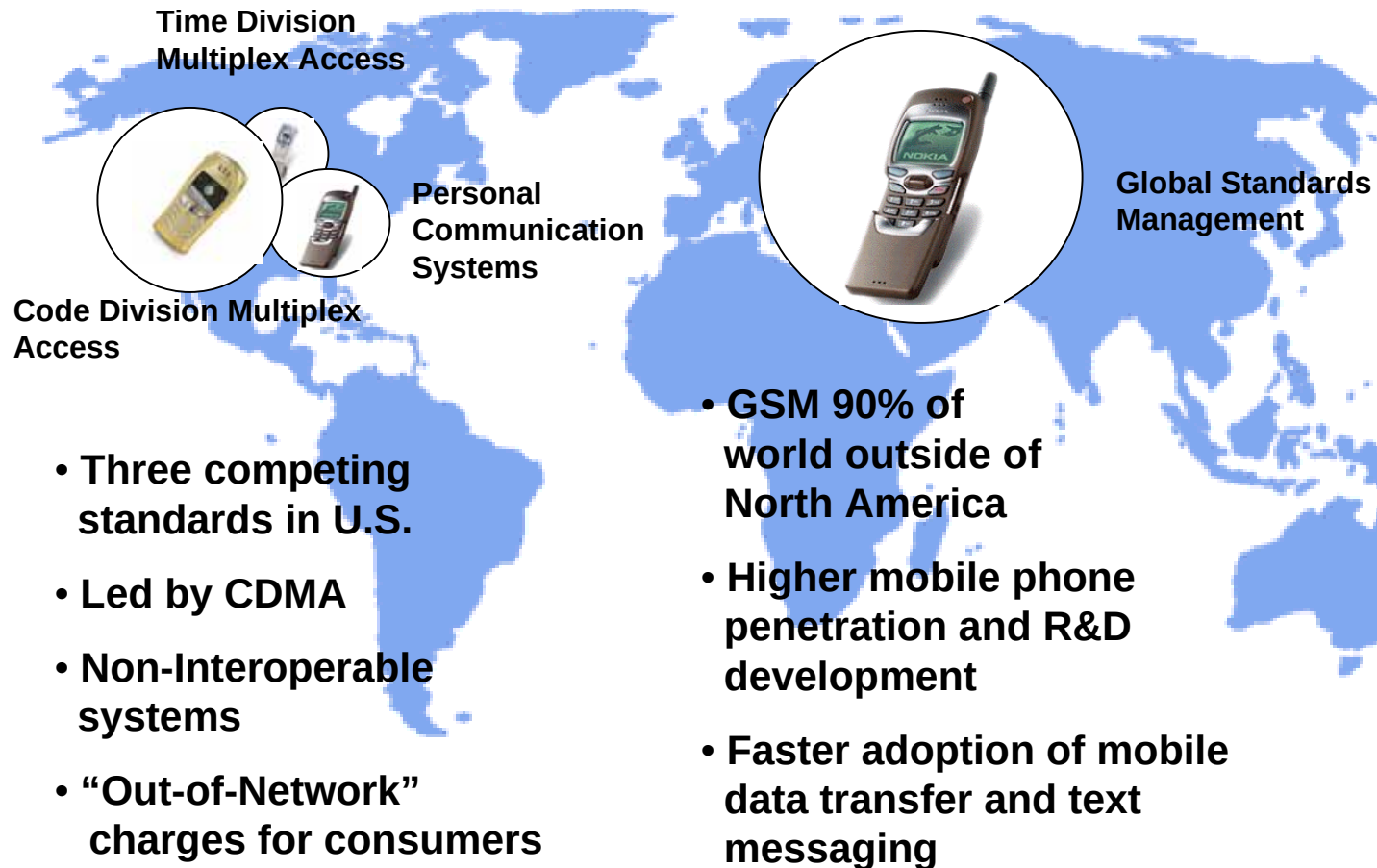
ISO

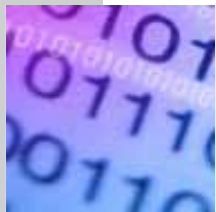
- Gen2 has been submitted to ISO
- ISO support is important in many parts of the world
- 18000 6-c passed first ballot
- Ongoing success for EPCglobal depends upon:
 - Being neutral and not for profit
 - Being user driven and global
 - Commitment to royalty free, open standards



Problem with Competing Standards

Pitfalls of Mobile Communications

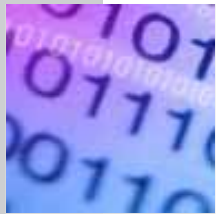




EPCglobal Overview

Standards Development Scope

- Technical Standards
 - Purpose: Specifications for physical RFID equipment (Gen 2)
 - Examples: Frequency; Chip Performance/Capability; Tag/Data
 - Built to work with, not compete with, application standards
 - Standards will graduate to ISO once complete
- EPCglobal Manager Number Registration
 - Purpose: Global registry for product classification systems
 - Examples: GTIN, SSCC, GRAI, GLN, VIN, CAGE, DODAAC etc.
 - Preservation of existing investments in enterprise technology and trading community collaboration efforts
- EPCglobal Network
 - Purpose: Network to create visibility for EPC-tagged items
 - Examples: Track and trace; authentication/pedigree, anti-counterfeit/diversion; recall/expiration, warranty/service
 - Tags are “license plates”; most information held in the network



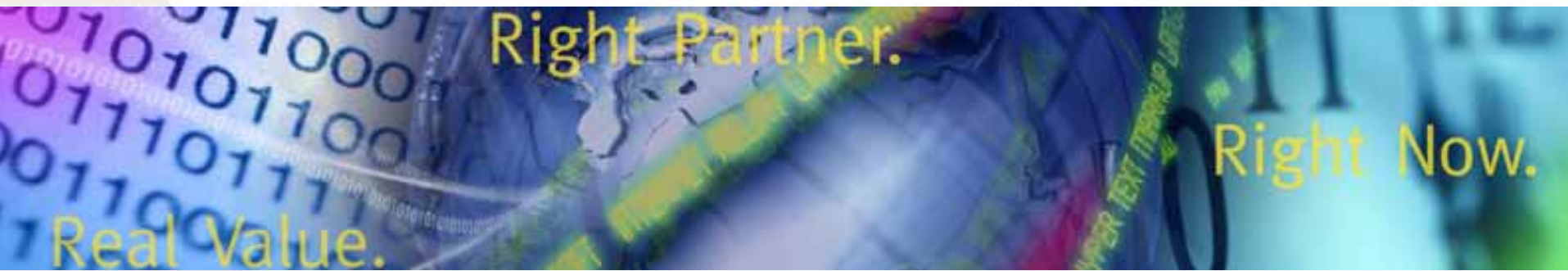
Value of RFID in Automotive

- Asset Tracking
- Real Time Location
- Anti-Counterfeiting
- Improved JIT manufacturing and replenishment
- Finished Goods tracking
- Reduced Cycle times



Advantages for the industry

- Common user defined global standards can improve a specific industry's efficiencies and reduce costs
- Efficiencies can be extended into support industries such as transportation, warehousing and customs
- Techniques applied in one industry can sometimes be successfully ported across to unrelated industries
- Wider use of standards leads to higher volumes which in turn decreases tag and equipment costs



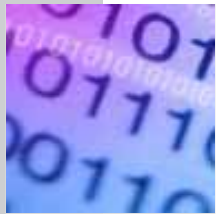
The Potential of EPC for the Automotive Industry

Patrick Schmitt, Dipl.-Wirtsch.-Ing.

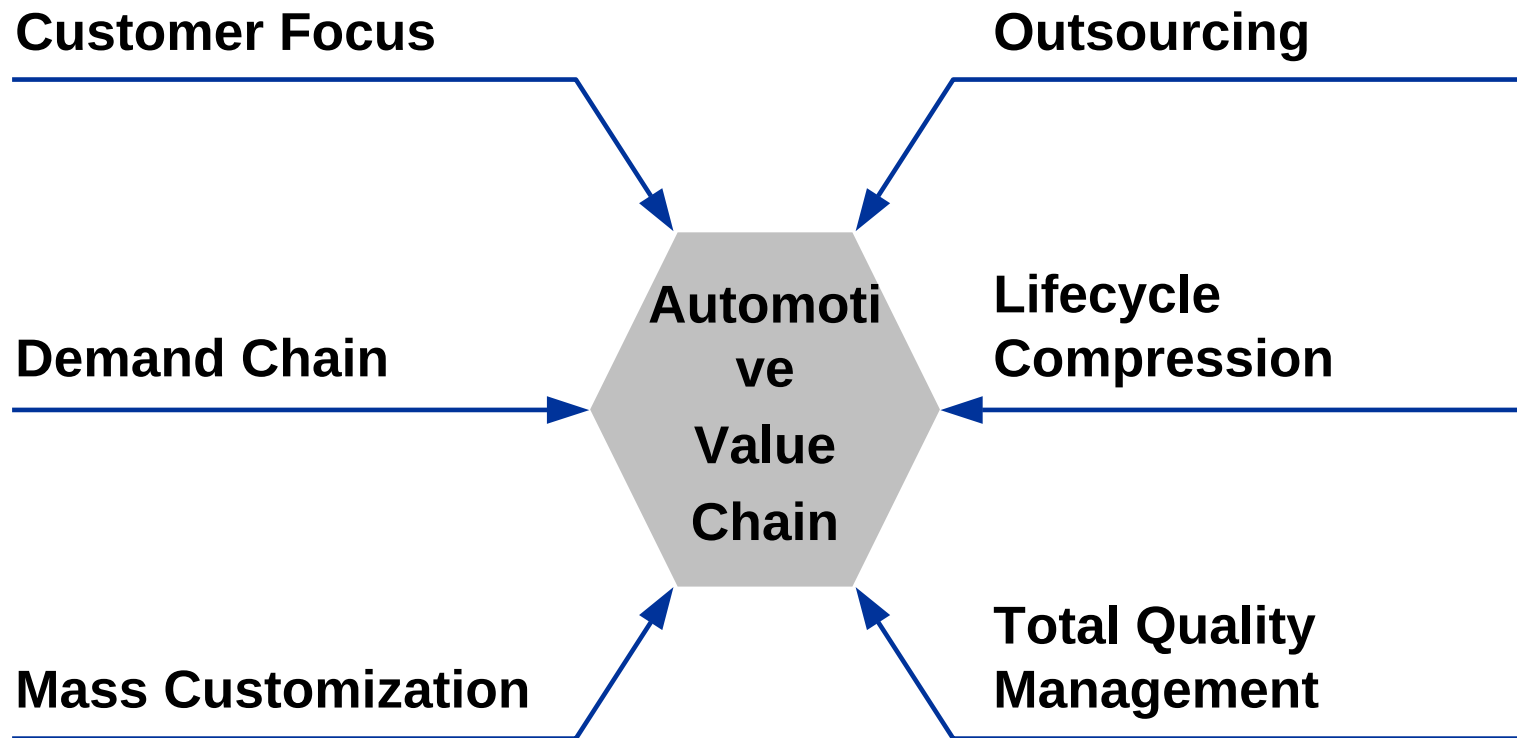
Dr. Florian Michahelles, Dipl.-Inf.

Swiss Federal Institute of Technology (ETH Zurich)

Dept. Management, Technology and Economics



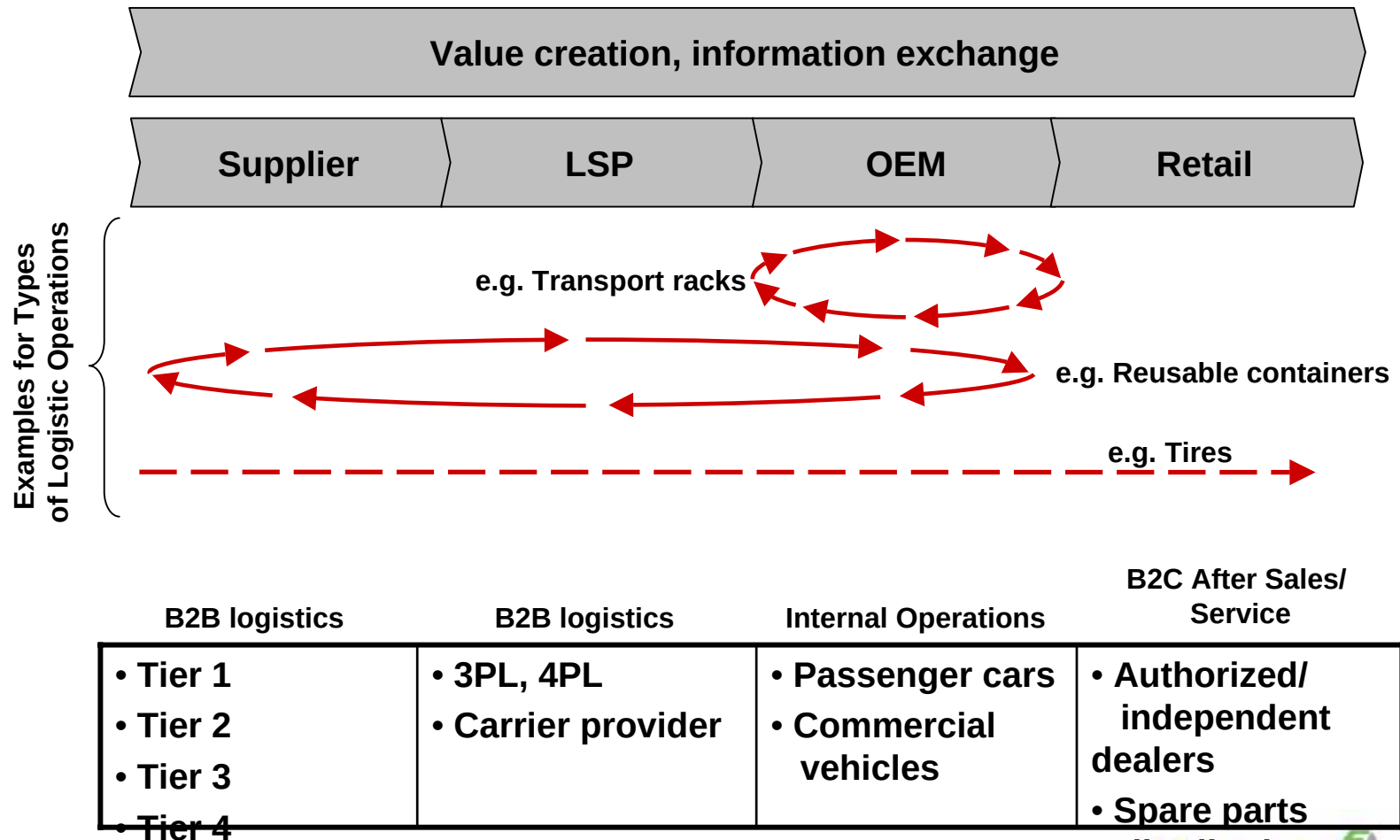
Strategic Challenges of the Automotive Industry

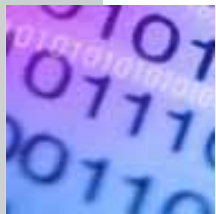


Strategic Trends in Supply Chain Management

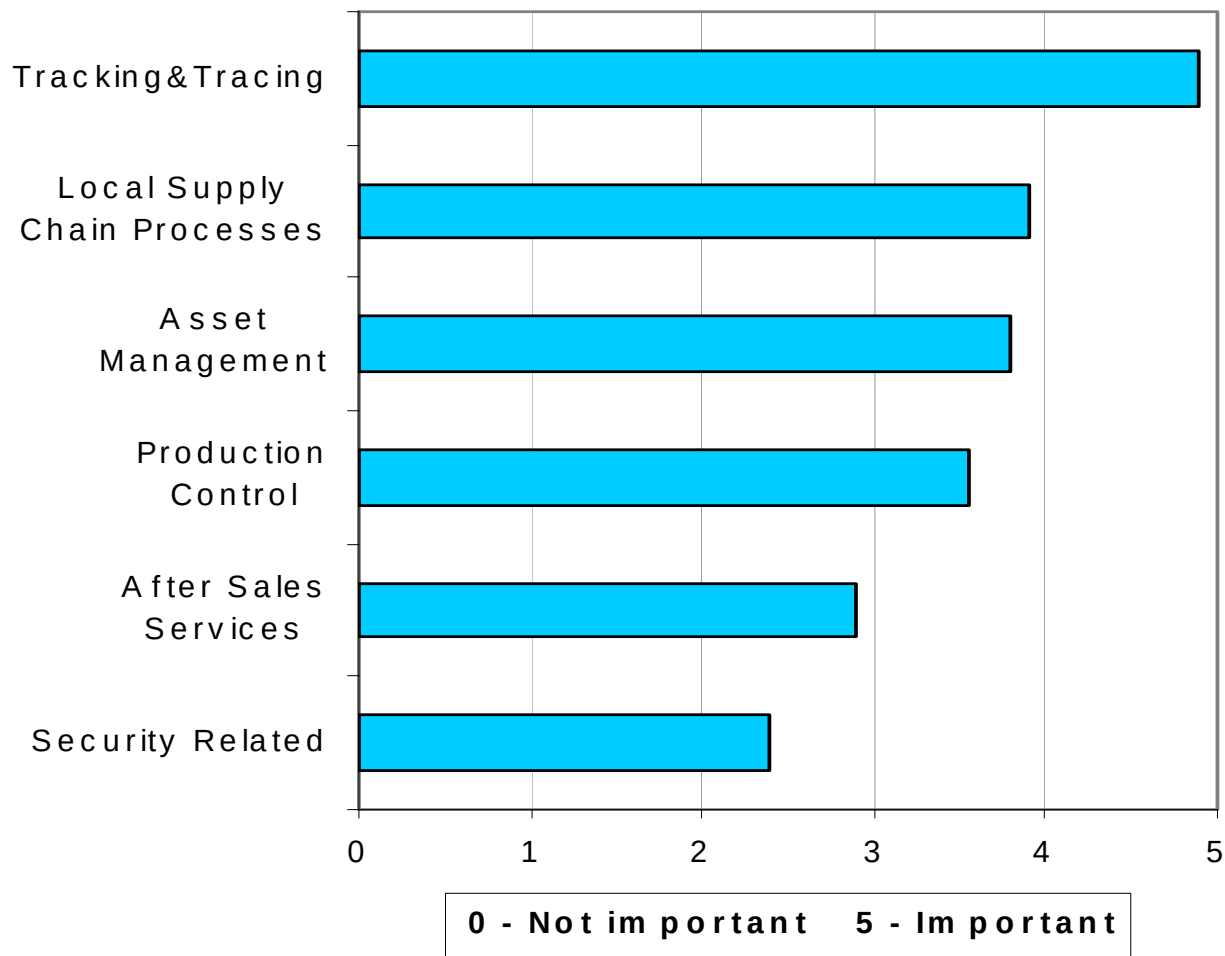


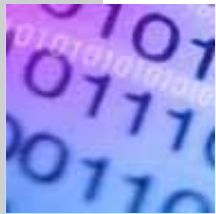
Automotive Value Chain





Priority of RFID Applications – Survey Results





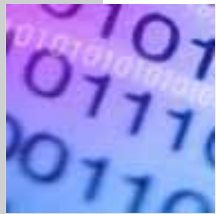
Drivers and Risks of an RFID/EPC enabled SC

Drivers

- Operational efficiency
- Strategic need
- Collaboration
- Advantages over barcode
- Handling efficiency
- Data granularity
- Technology leader

Risks

- Business case
- Organization
- Technology
- Integration
- High level of process reliability
- Standards



Concerns and Potentials regarding EPC Standards

Concerns

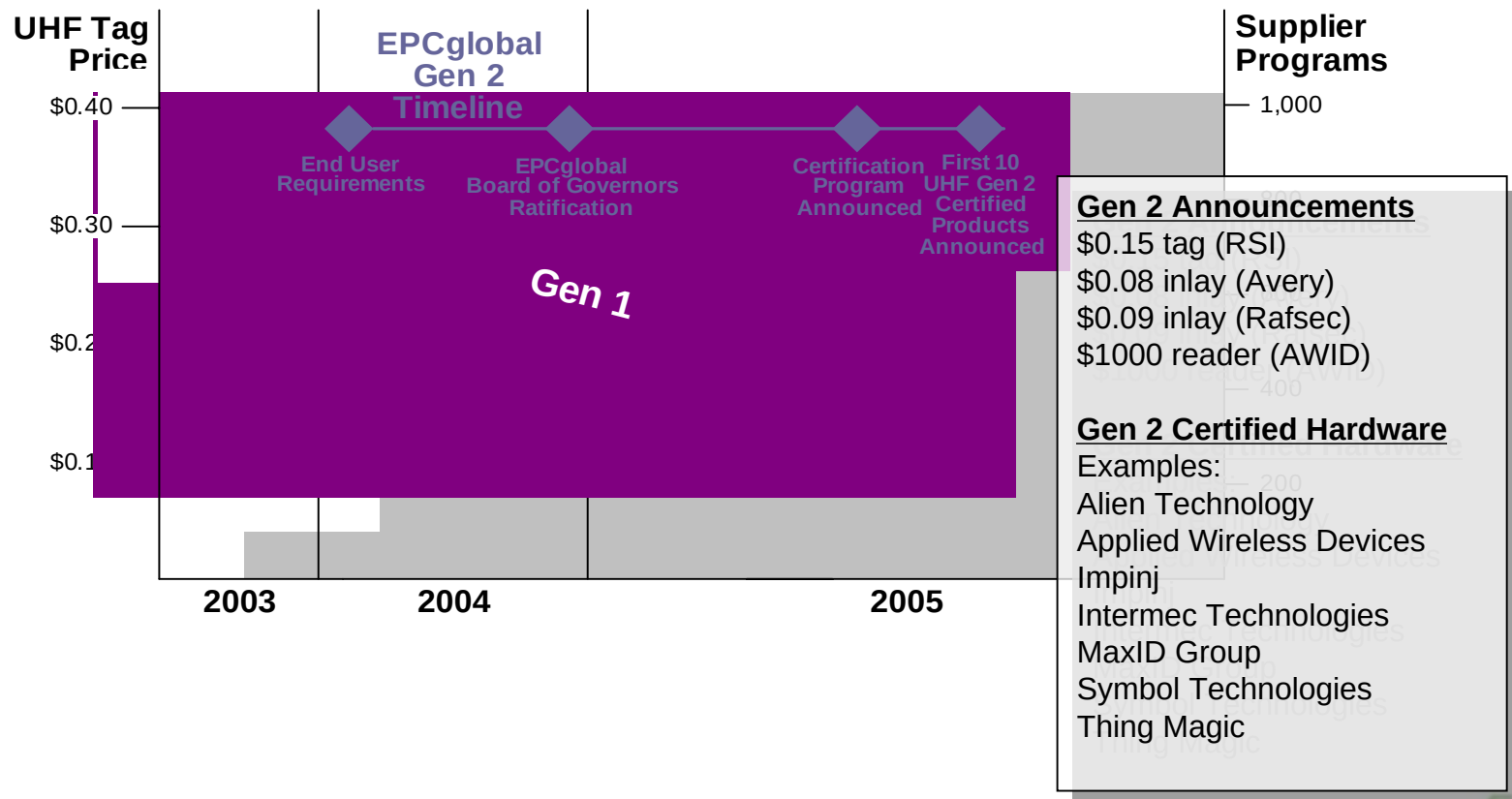
- EPC license fee and annual costs are too high
 - **Questionable compared to proprietary development**
- EPC is improper for closed loop systems with reusable containers
 - **Possibility to link EPC with additional information, date and time**
- Tags do not provide enough memory capacity for desired pack of data
 - **True for Class 1 BUT not true for Class 2 and following**
- No global standard regarding used UHF Frequencies
 - **EPC is the ONLY global standard**

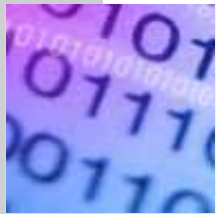
Potentials

- EPC standard will speed up RFID implementation
 - **Observer and late adopter will follow**
- EPC standard will lead to decreasing implementation costs
 - **BUT only prices of EPC tags will drop**
- Possible market infiltration with niche solutions
 - **Example: VDA-Material Tag is usable for genuine pallets only**

Competition in the hardware market

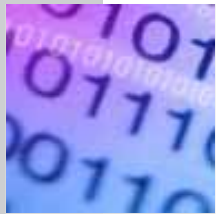
- Hardware prices continue to plunge





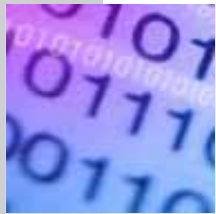
Current Business Action Groups

- FMCG
 - Implementations underway
 - Focus migrating to item level tagging
- Health & Life Sciences
 - Focus on supply chain pedigree
 - Pilots will be in US
- Transportation & Logistics
 - BAG formed late last year
 - Workgroup structure in place
 - Includes Customs, Logistics, IATA



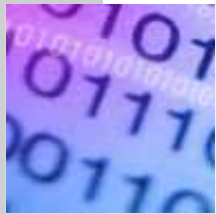
Bringing On New Industries

- First contact can be from industry, company or EPCglobal
- Initial sounding out of key industry players
- Formation of Discussion Group to assess viability of forming Business Action Group
- Charter, resources, workgroups, co-chairs, objectives, timetable and deliverables proposed
- Proposal signed off by MO's and EPCglobal Board
- First two BAG meetings held
- Hand off to Standards Development team



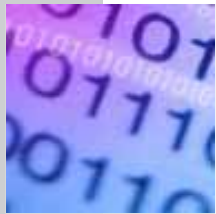
Automotive Summit (Discussion Group)

- 1st February 2006 at the RFID center in Bracknell (near Heathrow airport London)
- Attendance from: Daimler Chrysler, VW, Ford, Volvo, Saab, BMW, Michelin, Gefco, SKF, Bosch..
- Odette International, Odette Sweden, VDA, Gallia..
- GS1 Organisations: US, Germany, Japan, France, Sweden, Slovakia...
- See attached Program



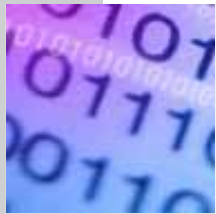
Purpose of Discussion Group

- Assesses potential benefits of implementing EPC Global standards and looks at requirements
- Made up of users and suppliers from the target industry
- Together with industry trade associations where appropriate
- Does not constitute a Business Action Group
- Membership of EPC Global not required to participate
- Facilitated by EPC Global staff
- May meet more than once



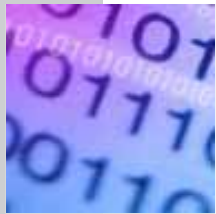
Pre-Requisites For Forming A BAG

- Majority of identified key industry players agree to participate
- Users willing to staff key positions such as Co-Chair
- Defined scope, charter, objectives and initial work plan
- Minimum overlap with existing or planned BAG's
- EPC Global President & Board sanction formation
- Sufficient EPC Global resources available for ongoing support



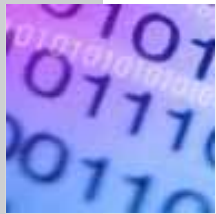
BAG Structures

- Leadership team composed of two Co-Chairs from the membership plus a facilitator from EPC Global
- Focus is on business requirements as technical development is thru Software and Hardware Action Groups
- Standards development is achieved thru the efforts of Work Groups which are formed as required to cover key business areas
- BAG members opt in to each Work Group that they contribute to
- Members may contribute to more than one Work Group



Future Business Action Groups

- Aerospace & Defense
 - Commercial Aircraft
 - Military supply chains
- Food and Beverage
 - Livestock
 - Fruits/vegetables subject to disease/infestation
- High-Tech
 - Computer industry
 - Consumer electronics
 - Industrial electronics
- Chemical
 - Petrochemicals
 - Industrial chemicals



Questions?

- www.epcglobalinc.org
- Contact details EPC:
- Nicholas Fergusson: fergusson@epcglobalinc.org
- Ian Robertson: irobertson@epcglobalinc.org
- Auto ID Lab, ETH Zurich
- Patrick Schmitt: pschmitt@ethz.ch